

Hello everyone, today I'm going to talk about *Weighted Average Defuzzification*, a simple yet powerful method used in fuzzy systems to convert fuzzy results into a single crisp value.

大家好，今天我要介绍的是“加权平均去模糊化（Weighted Average Defuzzification）”，这是一种在模糊系统中非常实用的方法，用来把模糊输出转化为一个清晰的、具体的数值。

There are **five main parts** in my talk:

part 1. Core Concept

The **Weighted Average Method** is a **defuzzification technique** used to convert a fuzzy set into a specific crisp value.

Here, the fuzzy set (A) (the output of fuzzy inference) is the union of (M) fuzzy sets.

Main idea:

For each output fuzzy set (i.e., the values corresponding to the linguistic variables), take its maximum membership degree $\mu(x)$;

Use these maximum membership degrees as weights to perform a weighted average on the corresponding crisp values x ;

The calculation formula is as follows:

$$x^* = \frac{\sum [\mu(x_i) \times x_i]}{\sum [\mu(x_i)]}$$

Among them:

x_i

: The distinct values corresponding to each language variable (e.g., 60, 70, 80...)

$\mu(x_i)$

: The maximum membership degree corresponding to this value

It is suitable for **fuzzy systems whose output membership functions are symmetric**(seimai吹克) **and regularly distributed**.

The advantage of this method is its simplicity in computation, yielding results very close to the COA (center of gravity method) while requiring less computational effort.

part 2. Example Illustration

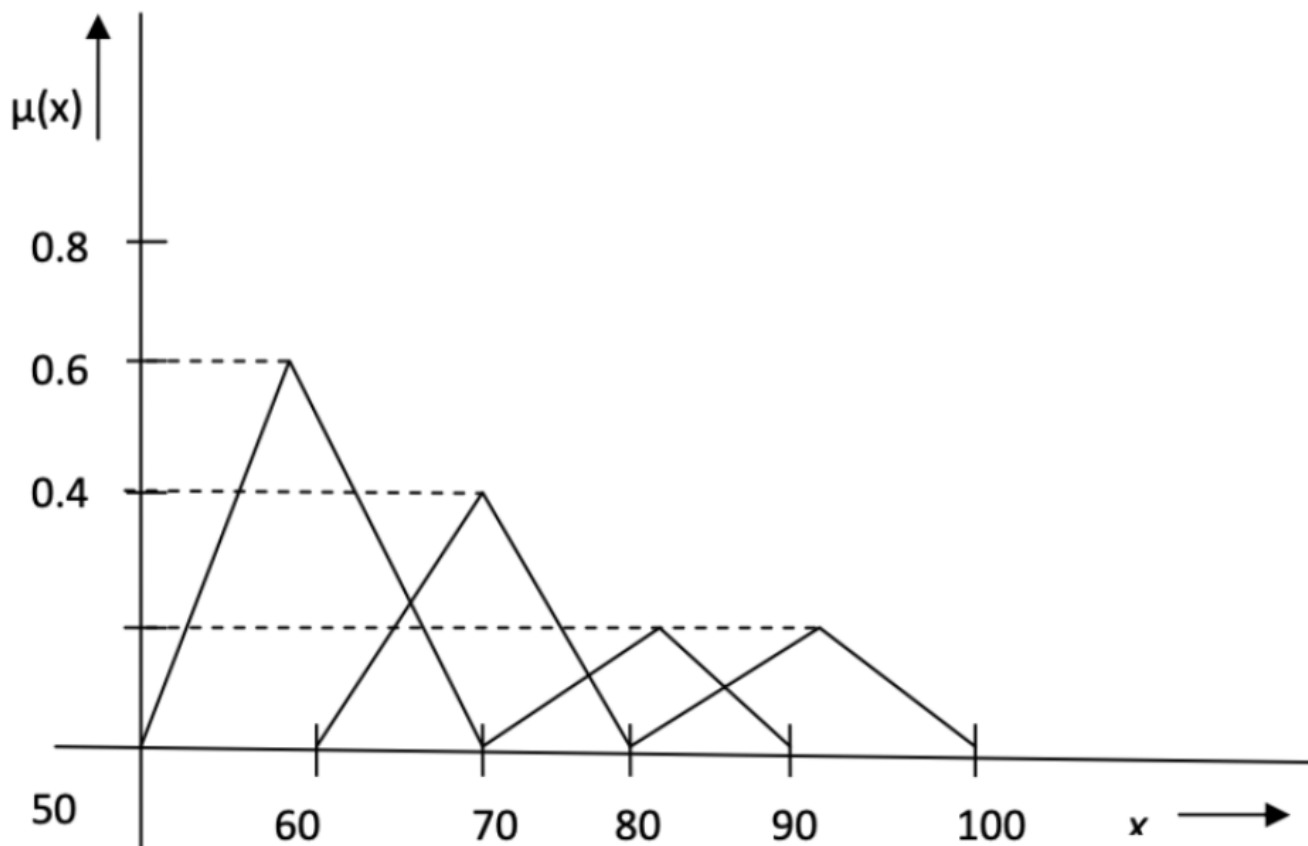
Let’s illustrate the Weighted Average Method through a simple example.
This is **not a complete fuzzy control system**, but rather an **example at the output layer**—we already have several fuzzy outputs (different grade levels and their membership degrees) and need to convert them into a **crisp score**.

Suppose fuzzy set (A) represents “*Student Performance Evaluation*”:
假设我们有五个模糊评价：及格（P）、中等（F）、良好（G）、优秀（VG）和卓越（E）。
Their peak values and maximum membership degrees are as follows:
它们的峰值和最大隶属度如下：

Linguistic Term	Meaning	x	$\mu(x)$
P	Pass	60	0.6
F	Fair	70	0.4
G	Good	80	0.2
VG	Very Good	90	0.2
E	Excellent	100	0.0

fuzzy set A

$$A = \{(P, 0.6), (F, 0.4), (G, 0.2), (VG, 0.2), (E, 0)\}$$



We design the output fuzzy set of “student grades” to be **regular, symmetric**(seimick), and **evenly distributed**, so that the defuzzification process is **more intuitive and easier to understand**, rather than fully representing a real-world case.

Applying the formula:

$$x^* = \frac{\sum [\mu(x_i) \times x_i]}{\sum [\mu(x_i)]}$$

$$x^* = \frac{60 \times 0.6 + 70 \times 0.4 + 80 \times 0.2 + 90 \times 0.2 + 100 \times 0}{0.6 + 0.4 + 0.2 + 0.2 + 0}$$

$$x^* = \frac{98}{1.4} = 70$$

Now, we multiply each x by its $\mu(x)$:

现在我们将每个 x 与它的 $\mu(x)$ 相乘:

$$60 \times 0.6 + 70 \times 0.4 + 80 \times 0.2 + 90 \times 0.2 + 100 \times 0.0 = 98$$

And the sum of $\mu(x)$ is 1.4.

So, $x = 98 \div 1.4 = 70$.

$\mu(x)$ 的和为 1.4，最后去模糊化结果为：

$x = 98 \div 1.4 = 70$ 。

That means, the final crisp value for the student's performance is seventy, corresponding to "Fair".

这意味着该学生的最终评分为 **70 分**，也就是“中等”水平。

part 3. Python Implementation

To implement this method, we can follow four simple steps in Python:

在 Python 中，我们可以按照以下四个步骤实现该方法：

1. **Generate membership functions** – create triangular fuzzy sets for each linguistic term.
生成隶属函数——为每个语言变量创建三角形模糊集合。
 2. **Extract maxima** – find the peak value and maximum membership degree of each set.
提取最大值——获取每个模糊集合的峰值和最大隶属度。
 3. **Apply the formula** – use the weighted average equation to compute the crisp output.
应用公式——利用加权平均公式计算最终输出。
 4. **Visualize the result** – plot fuzzy sets and mark the defuzzified result with a red line.
可视化结果——绘制模糊集合图，并用红线标出去模糊化结果。
- The code uses NumPy for calculation and Matplotlib for visualization.
代码使用 NumPy 进行计算，使用 Matplotlib 可视化。

Start with a helper function called **triangular()**.

This function takes four parameters — *a* (left point), *b* (peak point), *c* (right point), and *height* (the peak's maximum membership).

我们首先定义一个辅助函数 **triangular()**。

它有四个参数：左端点 *a*，顶点 *b*，右端点 *c*，以及峰值的最大隶属度 *height*。

Inside the function, we compute how each *x* in the domain(豆main) belongs to the fuzzy set.

The function forms a triangle by linearly (累naerly) increasing from *a* to *b* and then decreasing from *b* to *c*.

在函数内部，我们计算每个输入 *x* 对应的隶属度。

它通过从 *a* 到 *b* 线性上升，再从 *b* 到 *c* 线性下降，从而形成一个三角形的隶属函数。

This creates five fuzzy sets for the linguistic terms — **Pass, Fair, Good, Very Good, and Excellent**.

这样，我们就为五个语言变量——“及格、中等、良好、优秀、卓越”——生成了对应的模糊集合。

Next, for each fuzzy set, the program **scans (斯干) through all x values** to find where the membership function reaches its maximum.

接下来，程序会遍历每个模糊集合的所有 x 值，找到其**最大隶属度及对应的 x 值**。

We store these two lists:

- `x_values` : the peak positions of each fuzzy term
- `mu_values` : their maximum membership degrees

These (x, μ) pairs are essential inputs for the defuzzifier — they represent the **strongest evidence** from each linguistic term.

这些 (x, μ) 对是去模糊化计算的关键输入，代表每个语言变量的“最强证据点”。

Now apply the **weighted average formula**:

现在应用加权平均公式：

Here, each x_i is multiplied by its membership $\mu(x_i)$, and then divide by the sum of all $\mu(x_i)$.

In this example, the crisp result is seventy, which means the final defuzzified output is 70.

将每个 x_i 乘以其隶属度 $\mu(x_i)$ ，再除以所有隶属度的和。

最终得到去模糊化的结果 **70.0**，也就是“中等 (Fair)”水平。



part 4. Results and Visualization

the code uses **Matplotlib** to draw five triangular fuzzy sets from fifty to one hundred.

Each colored line represents one linguistic term.

最后，程序使用 **Matplotlib** 绘制出从 50 到 100 的五个三角形模糊集合。

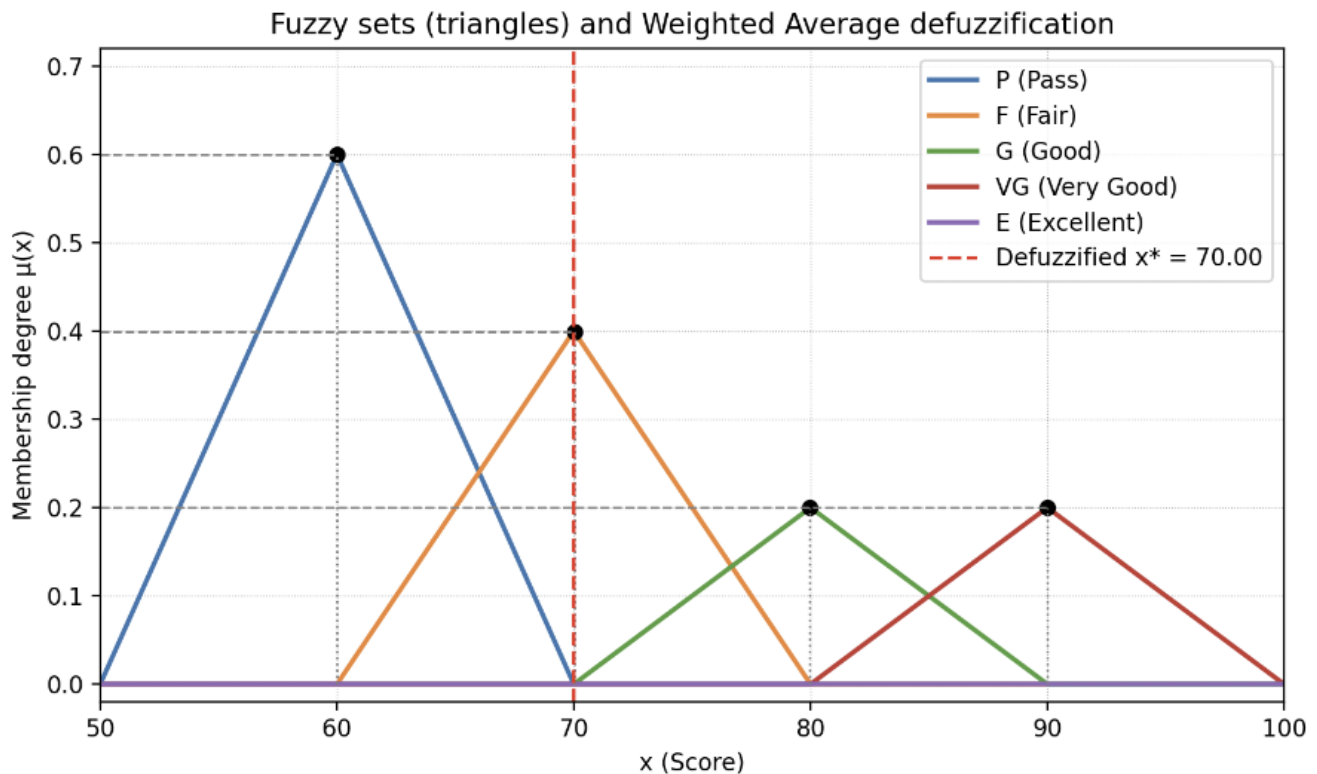
每条彩色线代表一个语言变量

Gray dashed lines mark each subset's peak and corresponding membership degree, and the **red dashed line** marks the defuzzified value.

helping us visually confirm that the weighted average result lies near the center of the overlapping regions.

灰色虚线显示了每个子集的峰值及其对应的隶属度,而**红色虚线**标出了最终的去模糊化结果。

从图中可以直观地看到加权平均结果 ($x^* = 70$) 位于重叠区域的中心位置。



🎯 part 5. Summary (撒莫瑞)

Let's summarize the Advantages and Disadvantages of the Weighted Average Defuzzification method.

我们来总结一下加权平均去模糊化方法的优缺点。

Advantages 优点:

- Simple and fast 简单且快速
- High accuracy for symmetric(sei麦吹克) sets 对称集合中结果准确
- Easy to understand and explain 易于理解和解释

Disadvantages 缺点:

- Not suitable for asymmetric(诶sei麦吹克) or irregular (一) membership functions 不适用于非对称或不规则的隶属函数
- Not a universal method 不是通用方案

In conclusion, the Weighted Average Method provides a **computationally efficient** and **conceptually intuitive** way to convert fuzzy information into crisp values — making it ideal for lightweight fuzzy control systems.

总的来说，加权平均法是一种**计算高效、概念直观**的去模糊化方法，非常适合用于轻量级的模糊控制系统中。

Thanks for listening.